Vladislav V. Vetroff

Dr. Bing Wang

CSE 4939W

5 November 2025

Assessing Ethical, Legal, and Commercialization Concerns

With every step in this project, from brainstorming to developing a prototype to pushing a product onto the market, there will always be factors and issues to consider. During the prototype development phase, there are concerns about the methodologies and approaches to developing the prototype; what technologies to use, how to use them to aid in the development of the project, etc. These concerns are important because they contribute to the time it takes to develop the project, as well as the outcome of the project. When it comes to delivering a finished product, however, real-world factors come into play. Questions arise, such as who the project is for, how to market the product, whether the project is safe and secure for its users, and what the legal implications are on the product's release, and many more. These questions ultimately concern the commercialization of the project and the legal and ethical implications of its commercialization. As the project stands, we are currently in the process of developing the prototype. Soon, as we transition from prototype to product, these concerns for commercialization and its ethical and legal implications will have to be addressed.

To understand how to address these concerns, we would need to state why these concerns relate to the project. In short, our project is a data collection and visualization solution for the UFSAE team's electric racecar. Our project will use OWL's proprietary hardware and software services, which include the "Quackerboard" (Ducks) and the Duck Management System (DMS).

In addition, our project will use their open-source networking protocol, the ClusterDuck Protocol, based on LoRa (Readme.Md, 2020). The terms of use for OWL's hardware and software services are stated and explained in their terms of service. According to OWL's Terms of Service, OWL is the "owner or the licensee of all intellectual property rights in our Services, including all source code, databases, functionality, software, website designs, audio, video, text, photographs, and graphics in the Services (collectively, the "Content"), as well as the trademarks, service marks, and logos contained therein (the "Marks") (Terms of Service, 2025)." The document elaborates further by stating that OWL's services "no part of the Services and no Content or Marks may be copied, reproduced, aggregated, republished, uploaded, posted, publicly displayed, encoded, translated, transmitted, distributed, sold, licensed, or otherwise exploited for any commercial purpose whatsoever, without our express prior written permission. (Terms of Service, 2025)." Unlike OWL's proprietary services, the ClusterDuck Protocol is open-source and under the Apache 2.0 license. The license allows the code for the ClusterDuck Protocol to be altered or distributed, whether commercially or not. The license also allows sublicensing, patent claims, and placement of warranties. However, the license states that "users cannot hold contributor(s) legally liable for any reason" and that it "does not grant you any rights to use the trademarks of the licensor (FOSSA Team, 2021)." In other words, we cannot use the ClusterDuck Protocol's likeness as our own, and we cannot sue OWL for any damages the ClusterDuck Protocol causes.

In addition to this information, it is important to consider that our primary customer for this project is the UFSAE team. Our project consists of training a machine learning model based on the data that the UFSAE team collected through their testing, which they have provided us with. As such, it seems that our project will be for the UFSAE team only, given the uniqueness

of the data set. This is our assumption going forward, as we cannot be sure that our project will be universal across all other FSAE teams or any other party that is interested in data telemetry on electric vehicles of the same caliber. We have doubts that our machine learning model could be applied to another car from a different FSAE team because we are not sure if other FSAE racecars are built like UConn's, given that our model will be specifically trained for the UFSAE team's racecar. By extension, we have doubts about the transferability of the project as a whole to other FSAE teams. However, while the project as a whole may not apply to other teams, this does not mean that the methodologies and systems behind this project are not beneficial otherwise. Other FSAE teams can use our methodology and system as a reference to suit their needs.

With the intellectual property protection on OWL's proprietary hardware and software, coupled with the fact that our project is uniquely tailored for the UFSAE team, it does not appear that we can release the project under the current circumstances. The insistence on the use of OWL's Quackerboard together with their DMS makes this not possible, unless we have been given explicit permission to do so (Terms of Service, 2025). However, the ClusterDuck Protocol is not limited, nor is it confined to the Quackerboards. According to the ClusterDuck wiki, OWL's Ducks could be recreated with alternative boards, as all it needs is "a microcontroller with a LoRa board, a power source, a data transfer cable, and an enclosure (optional) (Hardware, 2020)." These boards include: Heltec Lora ESP32 v2/v3, TTGO T-Beam SX127x, TTGO T-Beam SX126x, CubeCell ASR6501, and CubeCell GPS (Hardware, 2020). All that remains of OWL would be the DMS, which would either require the development of a similar cloud platform or be ignored in favor of the Ducks directly reporting to the web dashboard. Using alternative hardware and software to replace OWL's proprietary services would allow us to

open-source the project. Open-sourcing this project would allow the UFSAE team and other students at UConn to continuously contribute to the project's development long after its initial release. The only concern left here is the modularity of the project, where the user will be able to choose what to see on the web dashboard. For now, the dashboard's design and functionality are curated specifically for the UFSAE team.

There are ways to commercialize an open-source project. According to Morgan Lewis, an international law firm, these methods include publishing the project under a restrictive license, offering a premium version of the project, or publishing the majority of the project while keeping some parts of it proprietary (Semenikhina & Pennesi, 2023). All these methods are valid ways of commercializing an open-source project. However, we plan to open-source the project completely, for free. Therefore, commercialization of this project is not possible.

As for the ethical and legal issues regarding this project, there seems to be very little to address here. For starters, aside from the data that is collected from the electric racecar and the user's login information, there is no other data that the project collects. There is no point for this project to collect sensitive data, like payment or health information, as it ultimately does not need it. The project will be self-contained. Data that is collected from the racecar, be it from field-testing or during a race, would be stored and archived in the dashboard's internal database. In addition, the login information does not need to be sophisticated either; it could just be a single password or some private key only the UFSAE knows. The only information that the user has access to upon logging in would be the collected data from the racecar.

In conclusion, given the current configuration and circumstances of the project, it is very unlikely that commercialization will be possible because we will release the project as a free and open-source product. The project uses OWL's proprietary hardware and software, which

includes their Ducks and the Duck Management System. Alternative hardware and software would need to be considered and used to ensure that the project will be open source, with respect to their licenses. While open sourcing the project may present concerns for transferability between other FSAE teams' racecars, our methodologies and systems will nonetheless be beneficial for them.

Work Cited

"Clusterduck-Protocol/Readme.Md at Master · Clusterduck-Protocol/Clusterduck-Protocol." *GitHub*, 3 Mar. 2020, github.com/ClusterDuck-Protocol/ClusterDuck-Protocol/blob/master/README.md.

"Hardware." *GitHub*, 23 June 2020, github.com/ClusterDuck-Protocol/ClusterDuck-Protocol/wiki/hardware.

Semenikhina, Valentina, and Eric J. Pennesi. "Where's the Money? Options for Commercializing Technology, Part 1: Open Source Software." *Where's the Money? Options for Commercializing Technology, Part 1: Open Source Software –*, 26 May 2023, www.morganlewis.com/blogs/sourcingatmorganlewis/2023/05/wheres-the-money-options-for-commercializing-technology-part-1-open-source-software.

"Terms of Service." *OWL Integrations*, 28 Aug. 2025, www.owlintegrations.com/terms-of-service.

Team, FOSSA. "Open Source Licenses 101: Apache License 2.0." *FOSSA Blog*, FOSSA, 6 Feb. 2021, fossa.com/blog/open-source-licenses-101-apache-license-2-0/.